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Problem

- Directly fusing region features and words forces one multimodal encoder to discover alignment and interaction simultaneously.
- Web image-text pairs are abundant but captions can be incomplete or wrong, making hard one-hot targets unreliable.

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Momentum distillation

- An exponential-moving-average teacher produces soft targets for contrastive learning, matching and masked tokens.
- The teacher captures plausible alternatives and reduces overfitting to noisy web pair labels.

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Architecture

SEPARATE, ALIGN, THEN FUSE

- A ViT image encoder and a BERT text encoder first produce unimodal representations in a common space.
- A multimodal encoder then applies cross-attention from text to image features; no object detector or box labels are required.

ALBEF

Align before Fuse

Li et al. | arXiv:2107.07651v2

First align unimodal image and text representations contrastively, then fuse them with cross-attention while a momentum teacher softens noisy web supervision.

align then fuse

momentum teacher

vision-language

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Evidence

STRONG WITH ONLY 4M PAIRS

- ALBEF sets new results on retrieval despite using far less pretraining data than several competing systems.
- It improves VQA by 2.37 points and NLVR2 by 3.84 points over prior state of the art while enabling faster inference.

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Three objectives

ITC + ITM + MLM

- Image-text contrastive learning aligns global embeddings and supplies hard negatives for image-text matching.
- Masked language modeling reconstructs words from both modalities, grounding token prediction in visual context.

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Meaning & limits

GROUND BEFORE INTERACTION

- The align-before-fuse principle cleanly separates global correspondence from fine cross-modal reasoning.
- Momentum models and queues add training state; noisy web bias remains, and most downstream tasks still require fine-tuning.